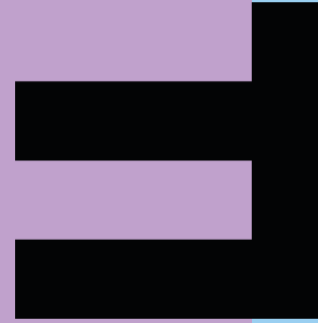


FHV

Vorarlberg University
of Applied Sciences



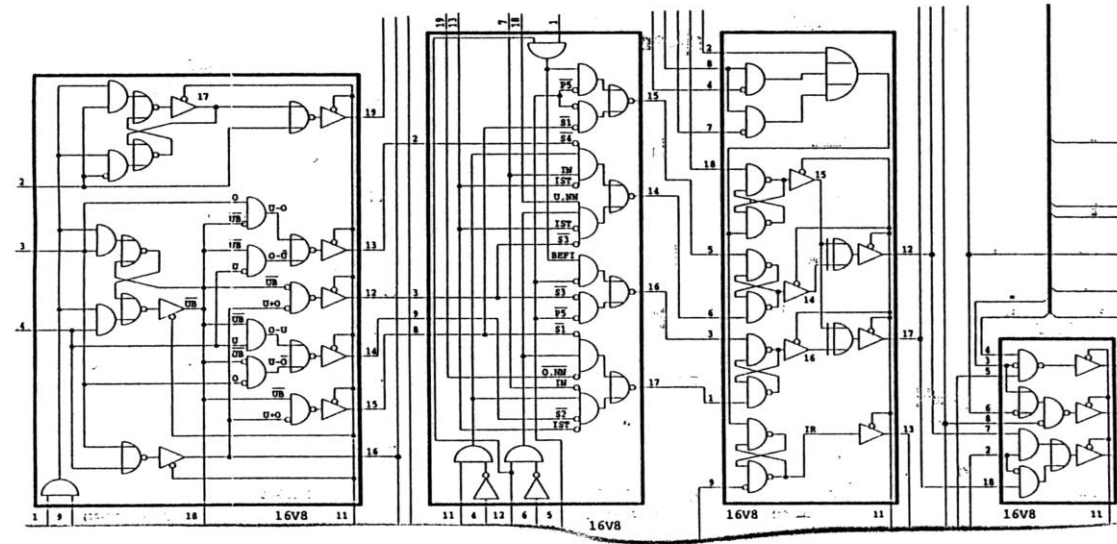
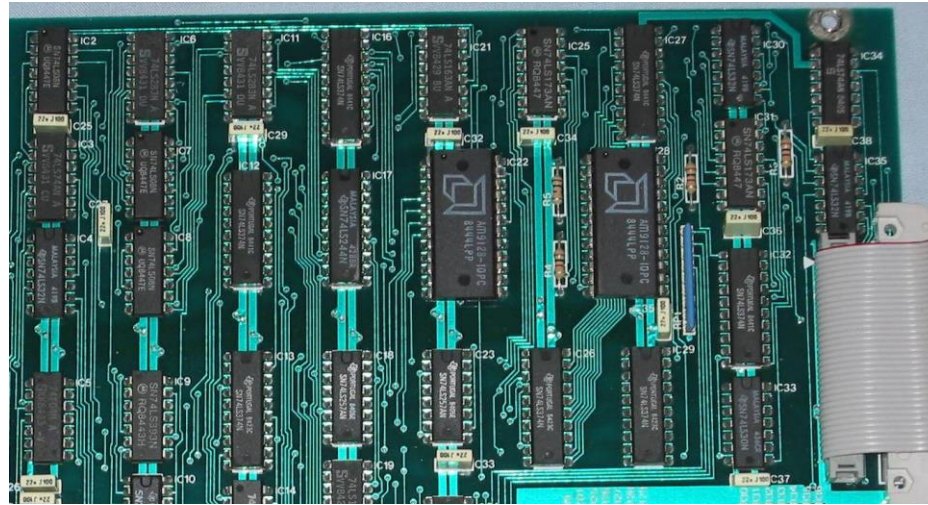
Programmable Logic

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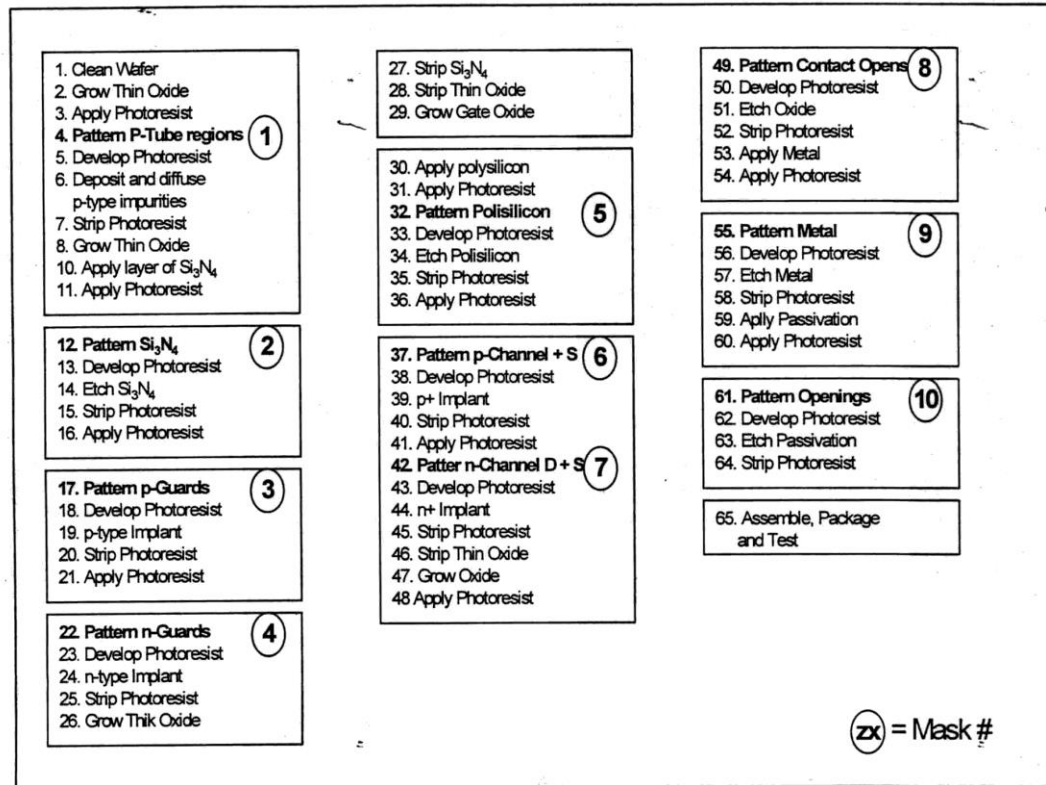
Horatiu O. Pilsan

Complex Digital Circuits



Costs in IC Manufacturing

- Masks needed in the manufacturing process of ICs:



- This is the most expensive part in manufacturing.



ASICs

ASIC = Application Specific Integrated Circuit

Pro and Cons of ASICs:

- High integration density
- High fixed (one-time) costs
- Low part costs
- Irreversible (one-time costs are lost if an error occurs...)
- High simulation (and verification) effort
- Long time to market (chip).



Full Custom IC

The entire layout of the IC is custom specific.

Pro and Cons of Full Custom ICs:

- All production masks are custom specific
- High fixed (one-time) costs (e.g. 100 k€)
- Highest density and complexity (usually up to 500 k equivalent gates)
- Lowest part costs (due to optimization)
- Longest time to market – up to 1 year.



Standard Cell IC

Manufacturer provides a library of standard cells which are used in the design.

Pro and Cons of Standard Cell ICs:

- All production masks are custom specific
- Slightly lower fixed (one-time) costs (e.g. 40 k€)
- Slightly lower density and complexity (usually up to 100 k equivalent gates)
- Slightly higher part costs (due to missing optimization)
- Slightly shorter time to market – a few months.



Gate Arrays

Manufacturer produces gate arrays (usually NAND) which are then connected according to customer needs.

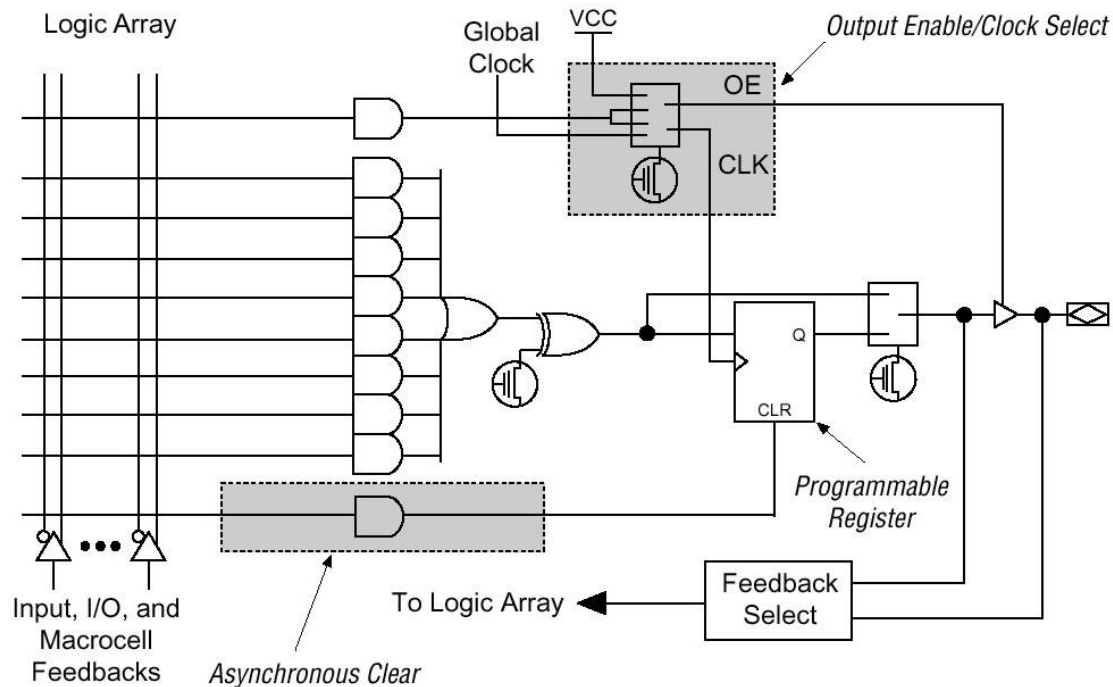
Pro and Cons of Gate Arrays:

- Only the last production masks (interconnect) are custom specific
- Lower fixed (one-time) costs (e.g. 20 k€)
- Lower density and complexity (usually up to 50 k equivalent gates)
- Higher part costs (due to missing optimization)
- Shorter time to market – a few weeks.



Simple Programmable Devices I

- Macrocell of a simple programmable device:



www.altera.com



Simple Programmable Devices II

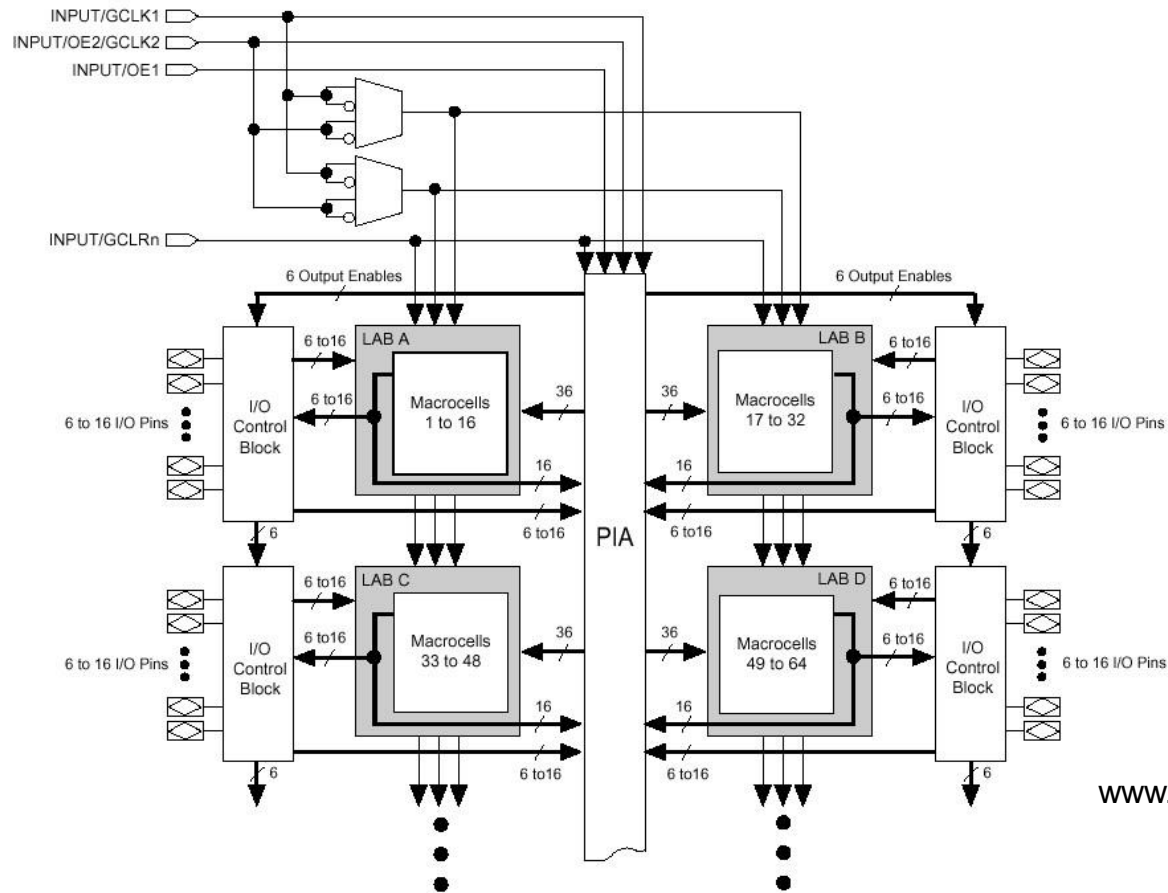
Typical characteristics of simple programmable devices:

- Most common type: GAL (Generic Array Logic)
- Macrocells (and outputs) 8 – 10
- IC case: 20 – 24 pin
- Technology: Flash
- Delay time: 5...25ns (pin to pin)
- Common parts: 16V8, 20V8, 22V10
- Dedicated input pins for Clear, Clock and Output-Enable.



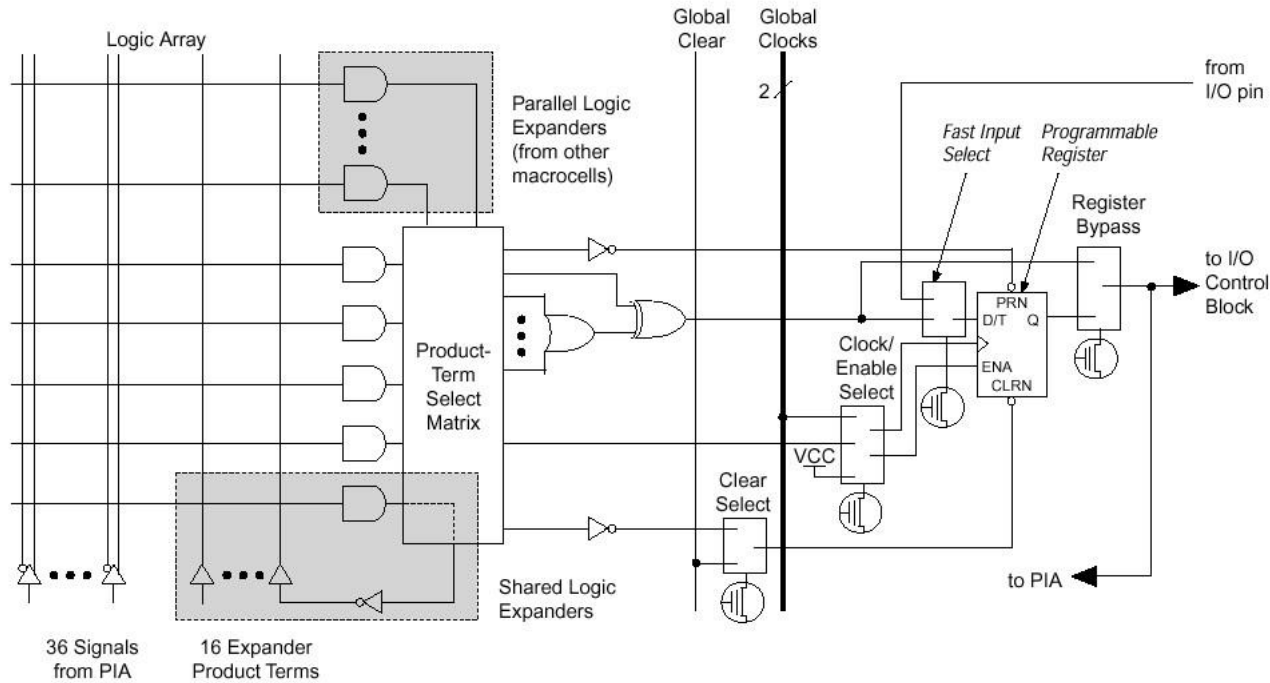
CPLD - Structure

- Structure of a **Complex Programmable Logic Device**:



CPLD - Macrocell

- **Macrocell of a Complex Programmable Logic Device:**



www.altera.com



CPLD - Characteristics

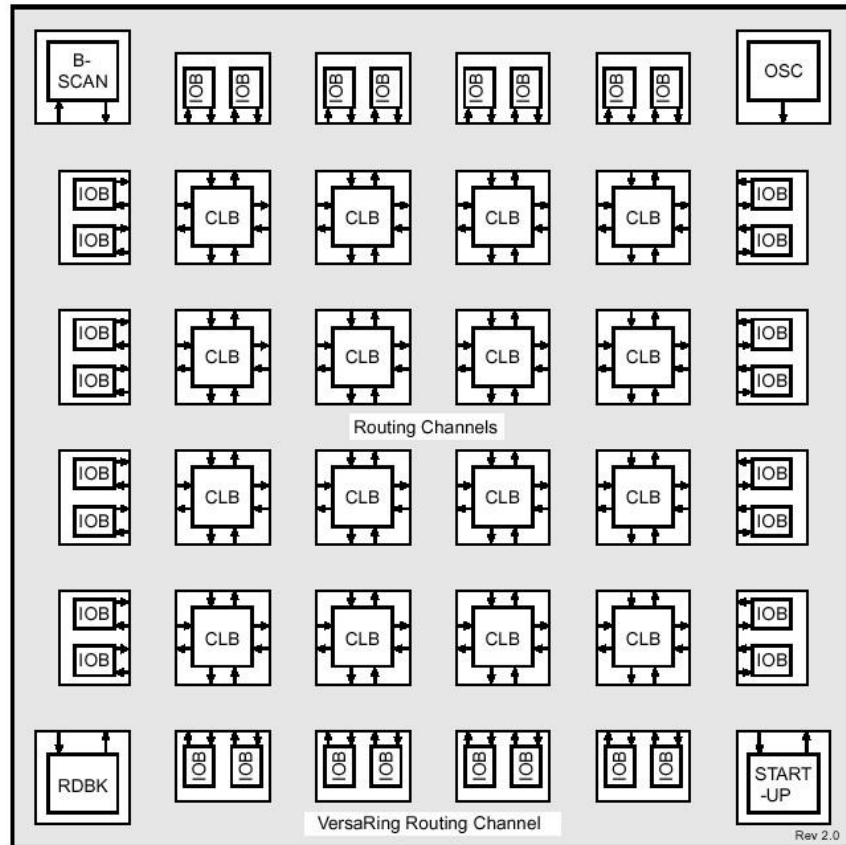
Typical characteristics of CPLDs:

- Macrocells: 32 – 500
- IC case: 44 – 356 pin
- Technology: Flash
- Delay time: 5...15ns (pin to pin)
- Equivalent gates: 1k – 10k
- Largest part of the device is the programmable interconnect array (PIA), which is the limitation, but which ensures fixed delay times.



FPGA - Structure

- Structure of a **F**ield **P**rogrammable **G**ate **A**rray:

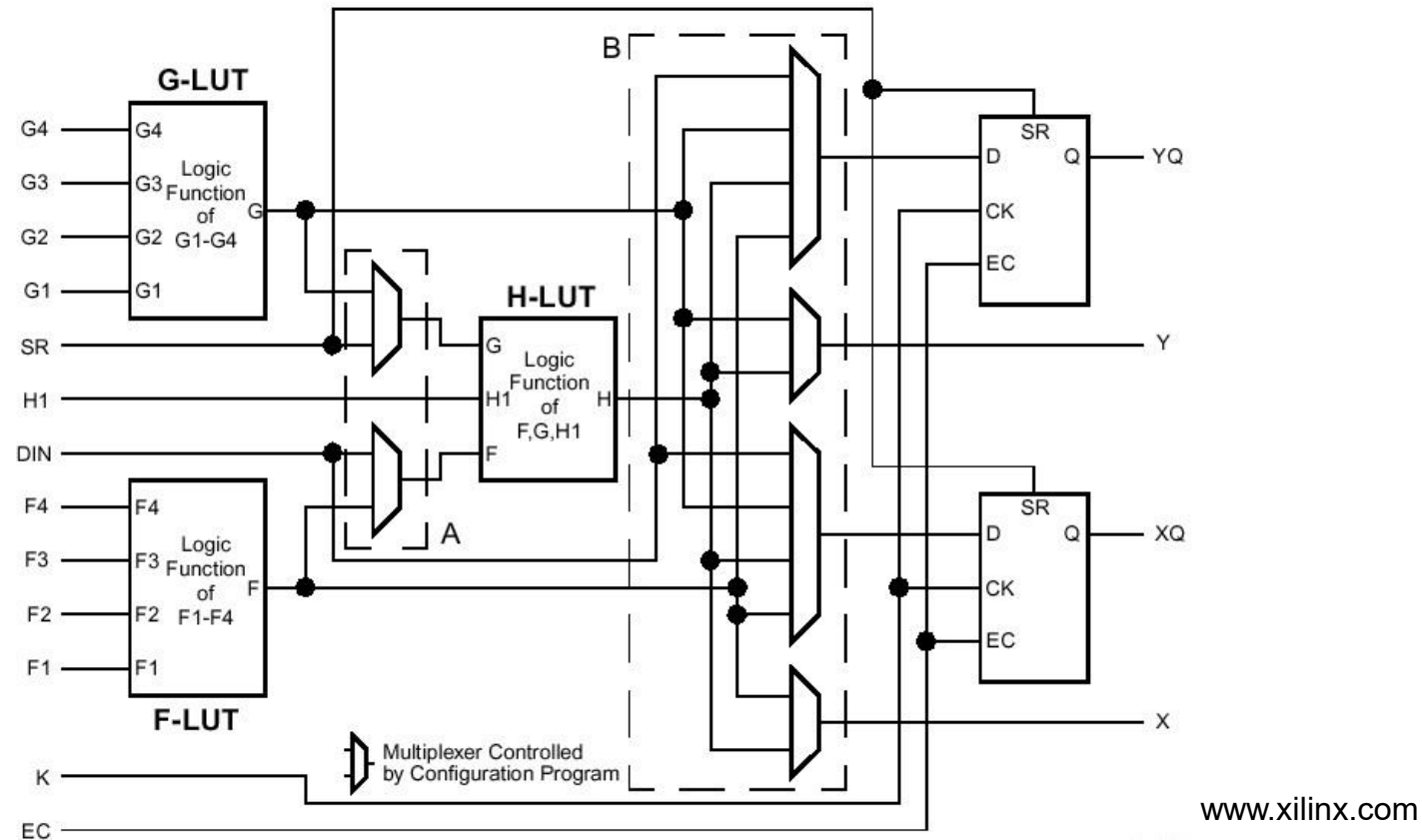


www.xilinx.com



FPGA - CLB

- Configurable Logic Block of a Field Programmable Gate Array:



FPGA - Characteristics

Typical characteristics of FPGA:

- CLB (sometimes called LAB) 100 – 50000
- IC case: 44 – 680 pin
- Technology: SRAM + (serial) Flash
- Delay time: can not be specified
- Equivalent gates: 10k – 100M
- Complex software needed for placement and routing of the FPGA.
- Configuration is loaded from a (serial) Flash into the FPGA at power-on.



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